

Literature Review Update Audit

Temporal, user-specific, evidence-conflict-aware, counterfactual minimal-change repair for weather-sensitive multi-day itineraries.

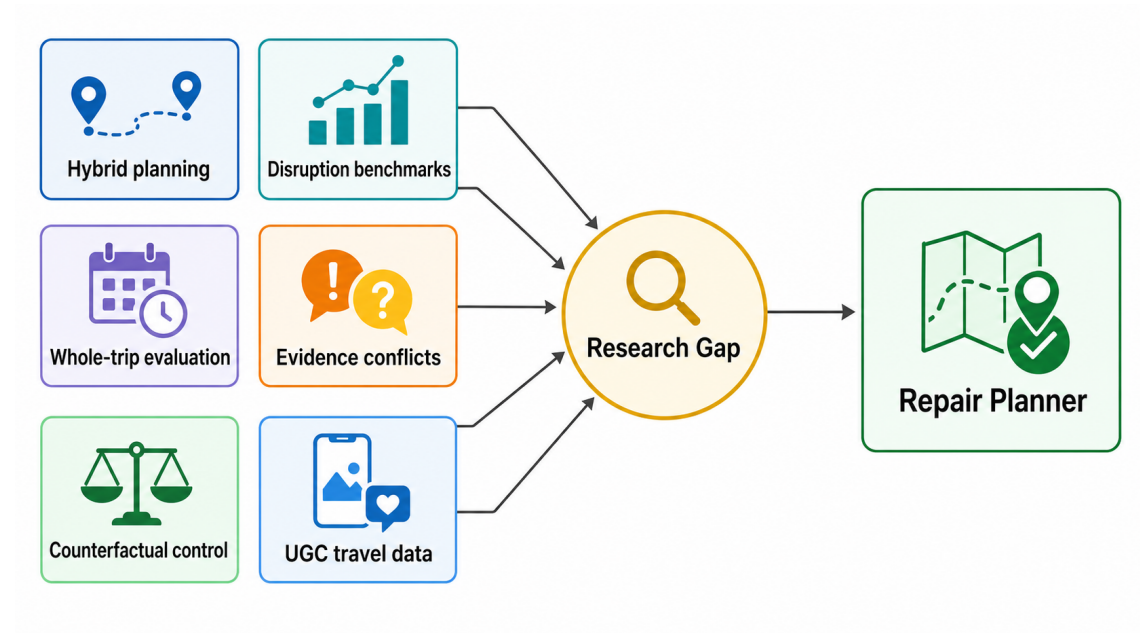


Figure 1. Conceptual gap map generated with the available imagegen skill; exact relationships are documented in the Markdown Mermaid source.

Date: 2026-07-02

Review type: scoped narrative literature review update, not a formal PRISMA systematic review.

Active skills followed: literature-review, literature-review-lean, plus imagegen and pdf where the review workflow required visual/PDF artifacts.

Review Objective

Update the project literature review so the publication framing is no longer "LLM plus optimizer" and is instead grounded as:

Temporal, user-specific, evidence-conflict-aware, counterfactual minimal-change repair for weather-sensitive multi-day itineraries.

The update must document the search scope, cite the closest novelty threats, synthesize them thematically, and identify what claims remain unsupported without experiments.

Search Strategy

Concepts Searched

- 1 LLM travel planning and language-to-symbolic optimization.
- 2 Itinerary modification, disruption revision, and minimal-change repair.
- 3 Whole-trip evaluation and tool-use travel benchmarks.
- 4 Conflict-aware retrieval and evidence-grounded planning.
- 5 Counterfactual and user-controllable recommendation.
- 6 Social-media/UGC travel information and route planning.
- 7 Exact-phrase checks for counterfactual/minimal-change itinerary repair.

Sources Checked

Source type / Coverage / Status

- **Source type:** Local PDF corpus | **Coverage:** Existing reference/* .pdf files and generated summaries | **Status:** Checked
- **Source type:** arXiv | **Coverage:** Current 2024-2026 travel-planning, RAG, and agent-planning preprints | **Status:** Checked by targeted web opens/search
- **Source type:** DOI/publisher metadata | **Coverage:** From Stay to Play DOI and PDF metadata | **Status:** Checked
- **Source type:** Local review docs | **Coverage:** related_work_outline, core_paper_reading_cards, deep report, addendum, evidence matrix | **Status:** Checked

Inclusion Criteria

- English-language peer-reviewed papers or preprints.
- Direct relevance to travel planning, itinerary recommendation, route optimization, travel-agent evaluation, disruption revision, evidence conflict, or counterfactual user control.
- Papers that threaten at least one broad novelty claim or support one component of the revised repair framing.

Exclusion Criteria

- Generic recommender or routing papers without travel/itinerary relevance.
- Product pages or blog posts without an associated paper.
- General LLM prompting papers without planning, retrieval, optimization, or travel relevance.

Citation Verification Log

Citation / Verification evidence / Use in review / Status

- **Citation:** iTIMO | **Verification evidence:** arXiv:2601.10609, title and abstract verified 2026-07-02 | **Use in review:** Itinerary modification; ADD/DELETE/REPLACE novelty threat | **Status:** Verified
- **Citation:** VeriTrip | **Verification evidence:** arXiv:2605.28683, title and abstract verified 2026-07-02 | **Use in review:** Multi-source contradiction and evidence-grounded travel-agent benchmark | **Status:** Verified
- **Citation:** TP-RAG | **Verification evidence:** arXiv:2504.08694, title, EMNLP reference, and abstract verified 2026-07-02 | **Use in review:** Retrieval-augmented spatiotemporal travel planning; noisy/conflicting references | **Status:** Verified
- **Citation:** RAG with Conflicting Evidence | **Verification evidence:** arXiv:2504.13079, title and abstract verified 2026-07-02 | **Use in review:** Generic conflicting-evidence RAG novelty threat | **Status:** Verified
- **Citation:** CONFACT | **Verification evidence:** arXiv:2505.17762, title and abstract verified 2026-07-02 | **Use in review:** Generic conflicting-evidence fact-checking/RAG novelty threat | **Status:** Verified

- **Citation:** COMPASS | **Verification evidence:** arXiv:2510.07043, title and abstract verified 2026-07-02 | **Use in review:** Multi-turn constrained optimization in travel-planning agents | **Status:** Verified
- **Citation:** CostBench | **Verification evidence:** arXiv:2511.02734, title and abstract verified 2026-07-02 | **Use in review:** Cost-optimal planning and dynamic adaptation in travel-planning tool-use agents | **Status:** Verified
- **Citation:** TravelBench | **Verification evidence:** arXiv:2512.22673, title, ACL 2026 note, and abstract verified 2026-07-02 | **Use in review:** Multi-turn, tool-use, and unsolvable travel tasks | **Status:** Verified
- **Citation:** TripCraft | **Verification evidence:** arXiv:2502.20508, title and abstract verified 2026-07-02 | **Use in review:** Spatiotemporally fine-grained travel-planning evaluation | **Status:** Verified
- **Citation:** ITINERA | **Verification evidence:** arXiv:2402.07204, title, related EMNLP DOI, and abstract verified 2026-07-02 | **Use in review:** LLM plus spatial optimization novelty threat | **Status:** Verified
- **Citation:** From Stay to Play | **Verification evidence:** Local PDF metadata and DOI 10.1016/j.apgeog.2016.10.002 verified 2026-07-02 | **Use in review:** UGC/social-media travel decision-support background | **Status:** Verified
- **Citation:** Braunhofer weather-aware tourism citation | **Verification evidence:** Existing deep-read note says source could not be reliably verified | **Use in review:** Removed from active citation backbone | **Status:** Quarantined

Updated Artifacts

Artifact / Update

- **Artifact:** docs/related_work_outline.md | **Update:** Added search methodology, inclusion/exclusion criteria, Mermaid gap map, AI schematic reference, and a new repair/evidence-conflict subsection
- **Artifact:** docs/project_literature_evidence_matrix.md | **Update:** Added rows for minimal-change repair operations, evidence-conflict handling, counterfactual repair explanations, and UGC/social-media travel evidence
- **Artifact:** docs/recent_papers_2023_2026_addendum.md | **Update:** Added iTIMO, VeriTrip, TP-RAG, ITINERA, TripCraft, TravelBench, COMPASS, CostBench, and From Stay to Play
- **Artifact:** docs/novelty_claim_verification.md | **Update:** Updated novelty-threat audit and safe contribution statement
- **Artifact:** docs/limitation_driven_itinerary_repair_method.md | **Update:** Reclassified From Stay to Play as verified background, not a core novelty citation
- **Artifact:** docs/chi_oriented_literature_review.md | **Update:** Replaced the unverified Braunhofer citation with verified From Stay to Play
- **Artifact:** docs/literature_review_slides.tex | **Update:** Replaced the unverified Braunhofer slide citation with Zhou et al. 2017
- **Artifact:** reference/1-s2.0-S0143622816306051-main_summary.md | **Update:** Added the missing paper-reading summary for From Stay to Play
- **Artifact:** reference/paper_summary_index.md | **Update:** Added From Stay to Play to the summary index
- **Artifact:** scripts/render_literature_review_audit_pdf.py | **Update:** Added a reproducible reportlab renderer for the audit PDF
- **Artifact:** output/pdf/literature_review_update_audit.pdf | **Update:** Generated the visual PDF artifact from this audit

Visual Artifact

The literature-review skill requires at least one visual schematic. The current session does not expose the separate scientific-schematics skill, so the available imagegen skill was used for an AI-generated schematic and the Mermaid source remains the exact reproducible diagram.

- AI schematic: docs/figures/literature_repair_gap_schematic.png
- Consuming document: docs/related_work_outline.md
- Reproducible text version: Mermaid block in docs/related_work_outline.md

Thematic Synthesis

Theme 1: Hybrid LLM + Optimizer Planning Is Established

TTG, TRIP-PAL, LLMAP, and ITINERA show that language-to-symbolic, language-to-graph-search, and LLM-plus-spatial-optimization travel planning are already active research areas. Therefore, the project must not claim novelty for combining an LLM with an optimizer.

Theme 2: Modification and Disruption Revision Are Established, but Mostly as Benchmarks

iTIMO formalizes itinerary modification through ADD, DELETE, and REPLACE perturbations. TripTide evaluates disruption revision and preservation/adaptability behavior. These papers make modification and disruption response central, but they do not provide the proposed solver/evaluator-backed, evidence-conflict-aware, counterfactual repair mechanism.

Theme 3: Evaluation Is Moving Toward Complete Plans

TravelEval, TripScore, TripCraft, TravelBench, COMPASS, and CostBench all push beyond surface plausibility. They motivate whole-trip metrics, tool-use boundaries, dynamic adaptation, and constrained optimization, but they do not replace the need for a repair method that reports hard feasibility, preservation, and counterfactual tradeoffs for a repaired route.

Theme 4: Evidence Conflict Is a Real Planning-Agent Problem

DRAGged into Conflicts, RAG with Conflicting Evidence, CONFACT, TP-RAG, and VeriTrip show that noisy, contradictory, and multi-source evidence cannot be hidden behind fluent outputs. The project should treat evidence-conflict handling as a required design element, not as a novelty claim by itself.

Theme 5: Counterfactual Control Exists in Recommendation, but Not as Itinerary Repair

User-controllable counterfactual recommendation and critique-based recommenders motivate "what must change" explanations. The project extends this idea to constrained multi-day route repair: what threshold must relax, which stop must move, or which substitution preserves feasibility.

Claim Boundary

Safe claim after this update:

We introduce a limitation-driven itinerary repair formulation that converts confirmed user intent and frozen disruption evidence into auditable ADD/DELETE/REPLACE/MOVE/RELAX/KEEP operations, explicitly labels conflicting evidence, optimizes for minimal change under temporal and weather-sensitive constraints, and reports whole-trip feasibility plus counterfactual explanations.

Claims still requiring experiments:

- Repair preserves more original intent than full replanning or heuristic replacement.
- Evidence-conflict handling improves route choices under contradictory weather/closure sources.
- Counterfactual explanations improve user comprehension, control, or error detection.
- The method outperforms baselines on feasibility, risk reduction, utility retention, runtime, or lodging consistency.

Claims to avoid:

- LLM plus optimizer is novel.
- UGC/social-media travel data plus routing is novel.
- Evidence-conflict handling is new in general.
- The system guarantees real-world itinerary validity.
- The system is an autonomous travel agent.

Quality Checklist

Requirement / Evidence / Status

- **Requirement:** Search methodology documented | **Evidence:** `related_work_outline.md` and this audit | **Status:** Complete
- **Requirement:** Inclusion/exclusion criteria stated | **Evidence:** `related_work_outline.md` and this audit | **Status:** Complete
- **Requirement:** Thematic synthesis rather than only study-by-study notes | **Evidence:** `related_work_outline.md`, evidence matrix, and this audit | **Status:** Complete
- **Requirement:** Citation verification performed | **Evidence:** Citation verification log above; source links retained in review docs | **Status:** Complete for newly added claims
- **Requirement:** Unverified citation quarantined | **Evidence:** Braunhofer weather-aware tourism citation removed from active backbone | **Status:** Complete
- **Requirement:** Visual schematic included | **Evidence:** `docs/figures/literature_repair_gap_schematic.png` referenced in outline | **Status:** Complete with available tool
- **Requirement:** PDF generation attempted | **Evidence:** Beamer slide rebuild failed due missing `beamer.cls`; non-LaTeX audit PDF generated separately | **Status:** Partially complete

Remaining Caveat

The slide PDF could not be rebuilt in the local TeX installation because `beamer.cls` is missing. The slide source was updated and failed-build byproducts were restored. A separate audit PDF is generated from this report using the bundled Python/reportlab runtime so the literature-review update still has a PDF artifact.